Homework 5
Statistics 2103

Please submit your code, as well as output (including plots), for all questions.

1. Ecologists researching the effects of nitrogen fertilizer on plant growth conducted an experiment by randomly assigning different nitrogen dose levels to 26 plants and observing each plant's biomass (i.e. how large they grow to be).

   The file `plants.csv` contains the following variables for the plants in the experiment.

   nitrogen: nitrogen fertilizer dosage (milligrams)
   biomass: biomass of plant (grams)

   (a) Make a scatter plot of the data, with "nitrogen" on the x axis. Draw the least squares regression line on the scatter plot (using "nitrogen" as the predictor variable).

   (b) Based on the scatter plot, describe the association between nitrogen fertilizer and plant growth. Does it appear to be strong or weak? Positive or negative?

   (c) Suppose we want to test whether there is a positive linear association between nitrogen fertilizer dose and biomass. State an appropriate null and alternative hypothesis for this problem.

   (d) Report the p value for the test.

   (e) With a rejection threshold of $\alpha = 0.05$, what would your conclusion be? Describe what that conclusion means in the context of this problem.

   (f) Find a 90% confidence interval for the slope of the regression line, and describe its interpretation in context.

2. The file `hurricanes.csv` contains the following variables for 62 hurricanes that made landfall in the US in the last 25 years:

   pressure: central pressure in millibars (mb)
   windspeed: maximum wind speed in knots

   Answer the following questions using the least squares regression model you fit to this data in Homework 4 (i.e. with central pressure as the predictor and maximum wind speed as the response).

(a) Find a 95% confidence interval for the average maximum wind speed among hurricanes with central pressure of 959 mb.

(b) Find a 99% confidence interval for the average maximum wind speed among hurricanes with central pressure of 959 mb. Is this interval wider or narrower than the interval in part (a)? Is it centered at the same value?

(c) Find a 95% prediction interval for the maximum wind speed of Hurricane Isaias, a specific hurricane with central pressure of 959 mb. How does this prediction interval compare to the interval you obtained in part (a)?

(d) For what central pressure does the model predict a maximum wind speed of 115 knots?

(e) Find a 95% prediction interval for the maximum wind speed of a hurricane with the central pressure value you obtained in part (d). Is this interval wider or narrower than the prediction interval in part (c)?

3. The file `salmonella.csv` contains salmonella bacteria colony density measurements for samples of chicken meat held at various different temperatures. The variables in the data are:

   `temperature`: chicken temperature (°C)
   `salmonella`: salmonella density

   (a) Make a scatter plot of the data, with "temperature" on the x axis. Draw the least squares regression line on the scatter plot.

   (b) Based on the scatter plot, describe the association between temperature and salmonella density.

   (c) What is the R^2 value for the regression model you fit in part (a)? Give an interpretation of this value.

   (d) Do you think the regression model from part (a) is useful for predicting salmonella density? Why or why not?

   (e) Do you think temperature is useful for predicting salmonella density in chicken meat? Why or why not?